The Cauchy-Riemann Equations

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Abstract

The goal of this paper is to give a brief introduction to complex analysis by discussing holomorphic functions.

1 Differentiation

Recall that we say a function $f : \mathbb{R} \to \mathbb{R}$ is differentiable at a point x_0 if the limit

$$\lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

exists. If it does, we denote the above limit by $f'(x_0)$.

Note that any function $f : \mathbb{C} \to \mathbb{C}$ may be written in terms of its real and imaginary parts:

$$f(z) = f(x + iy) = u(x, y) + iv(x, y).$$

We can identify $\mathbb{C}$ with the plane $\mathbb{R}^2$ through the map $(x, y) \mapsto x + iy$, and this will be useful in a moment.

Suppose now that $f : \mathbb{C} \to \mathbb{C}$, and consider a point $z_0 \in \mathbb{C}$. If

$$\lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

exists, then we say that f is *holomorphic at* z_0.

The two definitions above are identical, except that h can be any complex number. That is to say, we require the above limit to exist, *regardless* of the manner in which we approach the point z_0.

Consequently, suppose that f is holomorphic at $z_0 = x_0 + iy_0$, and let $h = h_1 + ih_2$. First, suppose that h is real, $h = h_1$. Then

$$
\begin{aligned}
f'(z_0) &= \lim_{h_1 \to 0} \frac{f(z_0 + h_1) - f(z_0)}{h_1} \\
&= \lim_{h_1 \to 0} \frac{(u(x_0 + h_1, y_0) - u(x_0, y_0)) + i(v(x_0 + h_1, y_0) - v(x_0, y_0))}{h_1} \\
&= \lim_{h_1 \to 0} \frac{u(x_0 + h_1, y_0) - u(x_0, y_0)}{h_1} + i \lim_{h_1 \to 0} \frac{v(x_0 + h_1, y_0) - v(x_0, y_0)}{h_1}.
\end{aligned}
$$

But observe that the two limits in the last line of the above expression are just the definitions of the partial derivatives of u and v at the point z_0! We therefore have

$$
f'(z_0) = \frac{\partial u}{\partial x}(z_0) + i \frac{\partial v}{\partial x}(z_0).
$$

Now, suppose that h is purely imaginary. That is, $h = ih_2$. Then

$$
\begin{aligned}
f'(z_0) &= \frac{1}{i} \lim_{h_2 \to 0} \frac{f(z_0 + ih_2) - f(z_0)}{h_2} \\
&= -i \lim_{h_2 \to 0} \frac{(u(x_0, y_0 + h_2) - u(x_0, y_0)) + i(v(x_0, y_0 + h_2) - v(x_0, y_0))}{h_2} \\
&= -i \lim_{h_2 \to 0} \frac{u(x_0, y_0 + h_2) - u(x_0, y_0)}{h_2} + \lim_{h_2 \to 0} \frac{v(x_0, y_0 + h_2) - v(x_0, y_0)}{h_2} \\
&= -i \frac{\partial u}{\partial y}(z_0) + \frac{\partial v}{\partial y}(z_0).
\end{aligned}
$$

You might be able to see what's next: We equate the two results to obtain

$$
\frac{\partial u}{\partial x}(z_0) + i \frac{\partial v}{\partial x}(z_0) = -i \frac{\partial u}{\partial y}(z_0) + \frac{\partial v}{\partial y}(z_0).
$$

Comparing the real and imaginary parts, we see that

$$
\frac{\partial u}{\partial x}(z_0) = \frac{\partial v}{\partial y}(z_0) \qquad\qquad \frac{\partial v}{\partial x}(z_0) = -\frac{\partial u}{\partial y}(z_0),
$$

which are known as the Cauchy-Riemann equations.

In particular, if $f : \mathbb{C} \to \mathbb{C}$ is holomorphic on an open set $\Omega \subset \mathbb{C}$, which just means that f is holomorphic at every point $z_0 \in \Omega$, then

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \qquad\qquad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}.$$

It is not a surprising fact that the space of holomorphic functions on an open set $\Omega \subset \mathbb{C}$ forms a vector space. Additionally, the ordinary product, quotient, and chain rule formulas hold as with the results in single variable calculus that we are familiar with.

One remark is in order: Notice that we may write

$$f'(z_0) = \frac{\partial u}{\partial x}(z_0) + i\frac{\partial v}{\partial x}(z_0) = \frac{\partial f}{\partial x}(z_0)$$

and

$$f'(z_0) = -i\frac{\partial u}{\partial y}(z_0) + \frac{\partial v}{\partial y}(z_0) = \frac{1}{i}\left(\frac{\partial u}{\partial y}(z_0) + \frac{\partial v}{\partial y}(z_0)\right) = \frac{1}{i}\frac{\partial f}{\partial y}(z_0).$$

Hence, the Cauchy-Riemann equations may be expressed neatly as

$$\boxed{\frac{\partial f}{\partial x} = \frac{1}{i}\frac{\partial f}{\partial y}}.$$

It is useful to define the differential operators

$$\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} + \frac{1}{i}\frac{\partial}{\partial y}\right) \qquad\qquad \frac{\partial}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} - \frac{1}{i}\frac{\partial}{\partial y}\right).$$

These partial differential operators are sometimes known as the Wirtinger derivatives.

Using the definitions above, we have the following result, which enables us to think of the (first) Wirtinger derivative as the total derivative of f.

Theorem 1.1 *Suppose that f is holomorphic at z_0. Then*

$$\frac{\partial f}{\partial \bar{z}}(z_0) = 0,$$

and

$$f'(z_0) = \frac{\partial f}{\partial z}(z_0) = 2\frac{\partial u}{\partial z}(z_0).$$

Proof: As f is holomorphic at z_0, it satisfies the Cauchy-Riemann equation

$$\frac{\partial f}{\partial x}(z_0) = \frac{1}{i}\frac{\partial f}{\partial y}(z_0),$$

and rearranging gives the first claim immediately. For the second claim, observe

$$2f'(z_0 = f'(z_0) + f'(z_0) = \frac{\partial f}{\partial x}(z_0) + \frac{1}{i}\frac{\partial f}{\partial y}(z_0) = \frac{\partial f}{\partial z}(z_0).$$

Expanding f into its real and imaginary parts and invoking the expanded Cauchy-Riemann equations gives the second equality in the claim. ■

For the last result of this paper, let us recall a familiar result from multivariable analysis [Spi65].

Theorem 1.2 *Suppose that $f : \mathbb{R}^n \to \mathbb{R}^m$. Then the total derivative $Df(a)$ exists if all the partials $D_j f_i(x)$ exist in an open set containing a and if each $D_j f_i$ is continuous at a, where $1 \leq j \leq n$ and $1 \leq i \leq m$.*

A similar result holds for holomorphic functions, as written below[1].

Theorem 1.3 *Suppose $f = u + iv$ is a complex-valued function defined on an open set Ω. If $u, v \in C^1(\Omega)$ and satisfy the Cauchy-Riemann equations on Ω, then f is holomorphic on Ω and $f'(z) = \partial f/\partial z$.*

Before proving this theorem, we make a small deduction about what it means for a function to be holomorphic. The limit

$$\lim_{h\to 0} \frac{f(z_0 + h) - f(z_0)}{h} = f'(z_0)$$

implies that

$$\lim_{h\to 0} \frac{f(z_0 + h) - f(z_0) - hf'(z_0)}{h} = 0.$$

In particular, a function f is holomorphic at a point $z_0 \in \Omega$ iff there exists a $a \in \mathbb{C}$ such that

$$f(z_0 + h) - f(z_0) - ah = h\psi(h),$$

where ψ is a function defined for h sufficiently small, with the property that $\psi(h) \to 0$ as $h \to 0$. This is similar to the motivation provided for the differentiability criterion for multivariable functions $f : \mathbb{R}^n \to \mathbb{R}^m$.

With that, we are ready to attack the problem.

[1] This result is derived from Chapter 1, Theorem 2.4 of [SS03]

Proof: The C^1 assumption on u and v implies that we may write their respective Jacobians as the 1×2 matrices

$$J_u = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \end{pmatrix} \qquad\qquad J_u = \begin{pmatrix} \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix}.$$

Hence, we may write

$$u(x+h_1, y+h_2) - u(x,y) = \frac{\partial u}{\partial x}(h_1) + \frac{\partial u}{\partial y}(h_2) + \|h\|\psi_1(h),$$

and

$$v(x+h_1, y+h_2) - v(x,y) = \frac{\partial v}{\partial x}(h_1) + \frac{\partial v}{\partial y}(h_2) + \|h\|\psi_2(h),$$

where $\psi_1(h), \psi_2(h) \to 0$ as $h \to 0$, $h = h_1 + ih_2$. Observe that

$$\begin{aligned} f(z+h) - f(z) &= (u(x+h_1, y+h_2) - u(x,y)) + i(v(x+h_1, y+h_2) - v(x,y)) \\ &= \frac{\partial u}{\partial x}(h_1) + \frac{\partial u}{\partial y}(h_2) + \|h\|\psi_1(h) + i\frac{\partial v}{\partial x}(h_1) + i\frac{\partial v}{\partial y}(h_2) + i\|h\|\psi_2(h), \end{aligned}$$

and invoking the Cauchy-Riemann equations yield

$$f(z+h) - f(z) = \left(\frac{\partial u}{\partial x} + \frac{1}{i}\frac{\partial u}{\partial y}\right)(h) + \|h\|\psi(h),$$

where $\psi(h) = \psi_1(h) + i\psi_2(h) \to 0$, as $h \to 0$. Thus, f is holomorphic, and

$$f'(z) = 2\frac{\partial u}{\partial z} = \frac{\partial f}{\partial z},$$

as required. ■

References

[Spi65] Michael Spivak. *Calculus on Manifolds*. Mathematics Monograph Series. CRC Press, 1965. ISBN: 9780805390216.

[SS03] Elias M. Stein and Rami Shakarchi. *Complex Analysis*. Vol. 2. Princeton Lectures in Analysis. Princeton, NJ: Princeton University Press, 2003. ISBN: 9780691113852.

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